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Optimal Post-Weld Heat Treatment Temperature in Aerospace Applications

1. Abstract

Advanced titanium alloys are used in aerospace manufacturing, where their low density, robust mechanical properties, and resistance to harsh operating conditions make them indispensable for critical structural components. Metallurgist and MIT graduate Stanley Abkowits developed this material at the Watertown Arsenal Laboratory, just a few miles from the Harvard Science and Engineering Complex (SEC) in Watertown, Massachusetts. However, the welding of this material introduces complexities not observed in more conventional metals. Their chemical reactivity at elevated temperatures, coupled with susceptibility to embrittlement when exposed to atmospheric gases, challenges the integrity of welded joints. In response, post-weld heat treatment (PWHT) processes have emerged as a strategy to restore or even enhance the mechanical characteristics of welded titanium by refining its microstructure, mitigating residual stresses, and improving hardness.

The project investigates the effectiveness of selected PWHT temperature profiles in modifying the hardness of a popular aerospace-grade titanium alloy. Unlike prior efforts focused primarily on recommended mid-range annealing conditions, this research explores a broad span of temperatures, extending both below and above conventional guidelines. By subjecting welded specimens to graduated thermal cycles and their surface and weld-region hardness, this work provides fresh insight into how subtle thermal

adjustments influence mechanical response. The objective is to refine the understanding of temperature selection. The findings encourage more nuanced approaches to thermal processing, supporting efforts to push the performance envelope of titanium-based aircraft components and increasing confidence in their long-term, in-flight durability.

2. Introduction

Titanium Grade 5 Alloy, Ti-6Al-4V, is an alpha-beta alloy considered to be the strongest of all titanium alloys. Alpha-betas alloys contain both the alpha and the beta phase, with aluminum being the principal stabilizer that strengthens the alpha phase and the principal beta stabilizer being vanadium, allowing the alloys to be hardened by heat treating and aging (Campbell 2008, pp. 533-533). Ti-6Al-4V is 6% aluminum and 4% vanadium. Ti-6Al-4V makes up 60% of the titanium used in aerospace applications and 80-90% used in airframe because of its high strength to weight ratio, heat resistance corrosion resistance and its high yield strength of 120,000 psi.

Post-weld heat treatment (PWHT) is a thermal process that is applied to welding in order to improve its mechanical properties. Titanium that is used in aerospace applications is heat treated to relieve its stress, improve resistance to brittle fracture, and increase its strength. The welded material goes through a process of heating, soaking, and cooling (Feng 2005). It is slowly heated to a temperature below its critical transformation temperature to avoid thermal shock, held at that temperature for a specific amount of time, and then gradually cooled. The temperature that the heat was held at depended on what temperature we were going to heat treat at because the furnace we used heats at a rate of 5 degrees C per minute. This process increases strengthening, hardenability, ductility, thermal stability, and improves stress relief.

According to a NASA Technical Memorandum, the optimal heat treatment temperature for Ti-6Al-4V is between 650-750 degrees C (Schmidt et al. 1966, pp. 21). The goal of this study was to determine which temperature of post-weld heat treatment would optimize the hardness of welded Ti-6Al-4V titanium alloy, using the optimal heat treatment temperature that NASA provided.

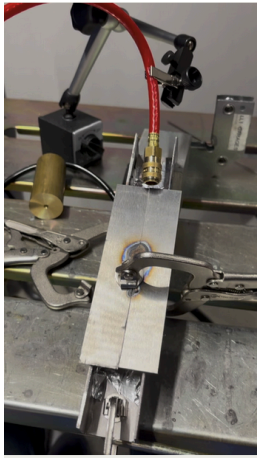
3. Experimental Methods

A 12" by 12" sheet of Ti-6Al-4V with a thickness of 1/16" was used for this experiment. A waterjet cutting machine was used to cut 1" by 3" sample pieces from the sheet. We then used argon-shielded Tungsten Inert Gas (TIG) welding to create 2" by 3" welded samples.

This study was composed of five experiments and one control, with a total of six welded samples. Using the heat treatment temperature of titanium that is recommended by NASA, we chose to test intervals of 200 degrees C temperatures. The five experimental samples were heat treated at different temperatures: 300 degrees C, 500 degrees C, 700 degrees C, 900 degrees C, and 1100 degrees C. The control sample was not heat treated. After treatment, the hardness of all six samples was tested and compared using the OMEGA-DIGI-RT-and-RST rockwell hardness tester.

TIG welding titanium is similar to welding stainless steel, except it's more challenging. It has high reactivity at elevated temperatures, so even when there is minimal air exposure, oxides and nitrides will form at the surface. Grade 5 also has a low thermal conductivity of 6.7 W/m·K, causing localized heat buildup to increase oxidation and deformation. Additionally, titanium has an oxide layer that fails at high temperatures. It leads to brittle oxides (TiO_2), nitrides (TiN), and hydrides (TiH_2).

Titanium alloy is also a material that can very easily be contaminated, especially above 650 degrees C, so a great deal of cleanliness and an auxiliary inert-gas shielding



must be used for welding (Donachie 1987, pp. 315). Acetone was used to disinfect the samples before TIG welding. At this temperature, it is extremely reactive with air and hence to prevent oxidation, an argon gas trailing shield and back purge was used.

Essentially, there was argon gas following, on, and under the weld.

If the shielding is not sufficient, an alpha case layer will form

(Phillips 2016, pp. 228) which is a brittle layer that is susceptible to

small cracks that can reduce mechanical properties. Various different set ups of the trailing shield and back purge were attempted, but ultimately the back purge was created using a 6" long metal square tube with the help of Rachel DeLucas. One surface of the piece of tube was removed, a hold was welded where the argon gas tube was to be inserted, and the sides were closed with steel foil.

Welding each sample took about 30 minutes. There was a lot of trial and error, as well as patience when it came to welding the six final samples that we used. The tungsten electrode continuously had to be grinded because econtact with the material would cause the electrode to be contaminated. We found the welded samples to have a lot of contamination post-weld, but we found less and less contamination as our welding improved. Different colors revealed different information about surface oxidation. A gold color indicated that the weld was protected down to 540 degrees C, and the contamination could be removed with a wire



brush (Donachie 1987, pp. 317). Blue indicated that the contamination was deeper and occurred after the metal was solidified, but it could be removed with a rotary file (Donachie 1987, pp. 317). The white, almost chalky color meant that the weld was very contaminated and happened when it was still liquid.

4. Results

The rockwell hardness test can be used to approximate a material's characteristics, such as rigidity, ductility, mealability, and fatigue. Scale B was used by the hardness tester to calculate the hardness of all of our samples. This scale used a 1.5875 mm Ball Indenter, with an initial test force of 98.07 N and main test force of 882.6 N. Each sample was tested at two points: one directly on the weld and one on the flat surface. For comparison, the Ti-6Al-4V sheet has a hardness of 85.6 HRB before any welding.

The hardness of our six samples surprised us (see Table 1). The hardness tester indicated that the welded sample that was post-weld heat treated at 300 degrees C had greatest hardness at the weld and at the source: 79.9 HRB at the weld and 101.3 HRB at the surface. The sample that was welded at the highest temperature, 1100 degrees C, had a hardness of 61.1 HRB at the weld. This temperature resulted in the least hardness.

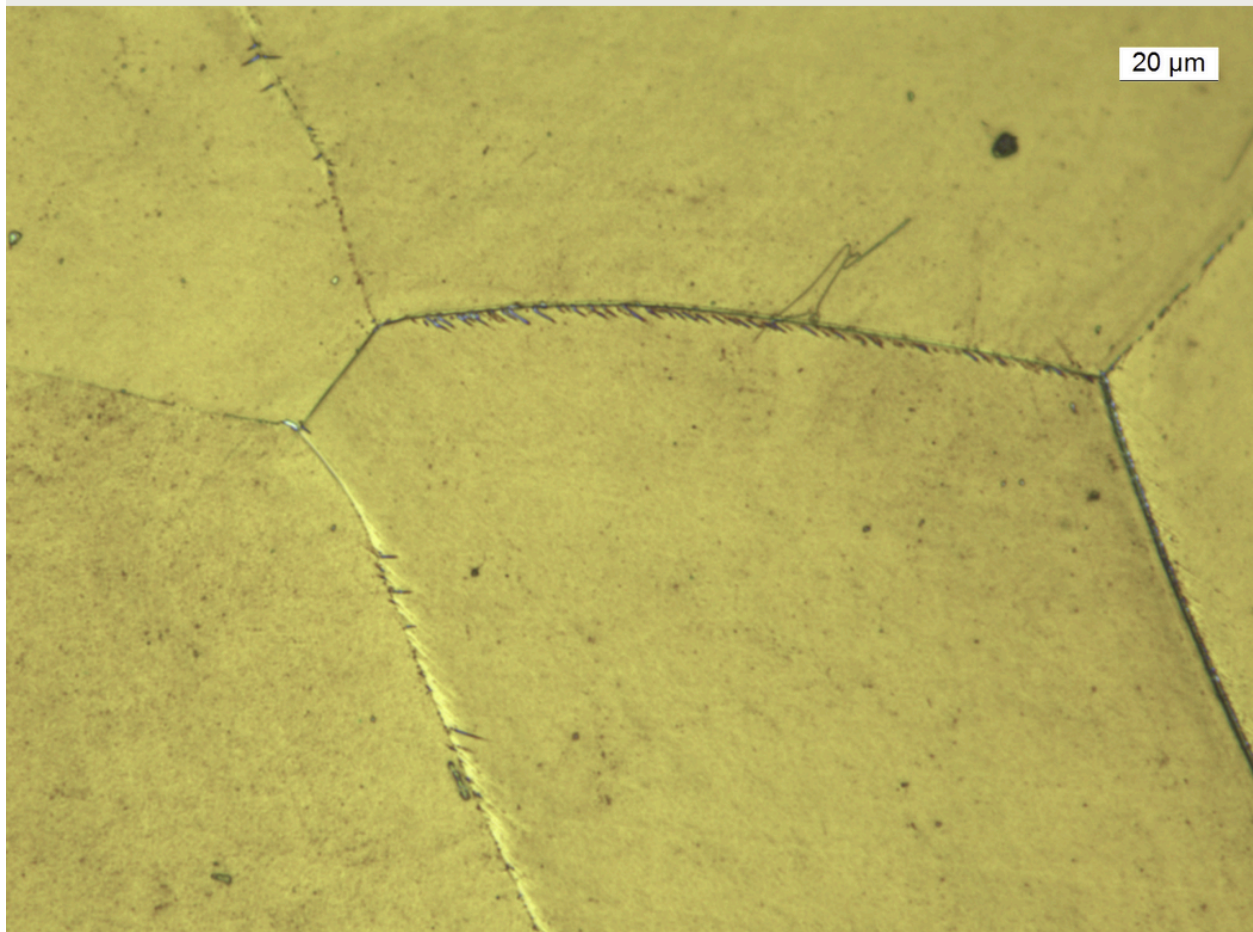


Table 1. Hardness at different post-weld heat treatment temperatures at the weld and at the flat surface of each sample.

PWHT Temperature	Hardness At The Weld	Hardness At The Flat Surface
No PWHT	79.0 HRB	78.2 HRB
300 C	79.9 HRB	101.3 HRB
500 C	78.0 HRB	82.9 HRB
700 C	84.6 HRB	90.9HRB
900 C	79.3 HRB	65.5 HRB
1100 C	61.1 HRB	81.8 HRB

		(unoxidized/clean surface) 52.3 HRB (oxidized surface)
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5. Discussion

Contrary to previously established recommendations, the results here suggest a significantly lower temperature threshold for post-weld heat treatment to achieve desirable hardness in Ti-6Al-4V welds. While earlier research had identified a considerably higher temperature as optimal, our testing indicates that exposure near 300°C produced the greatest improvement in surface hardness. Several factors may explain this discrepancy.

First, environmental conditions during thermal processing were not strictly controlled. Without a consistently inert atmosphere, elevated temperatures likely promoted unwanted reactions between the alloy and surrounding gases. These reactions could have created brittle surface layers or altered the microstructure in ways that reduced hardness at higher treatment temperatures. The formation of oxide layers or the entrapment of impurities may have been minimized at a lower temperature, allowing the alloy's intrinsic properties to remain more favorable.

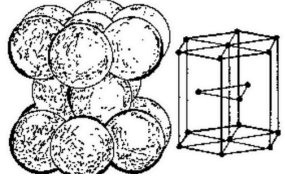
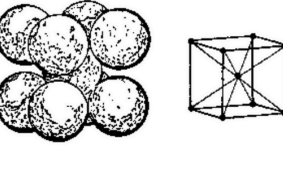
Second, the chosen heat treatment durations were not uniform because the rate of temperature increase was fixed. Each targeted setpoint thus experienced a different effective “soak” time, potentially creating variable diffusion profiles and phase compositions. This effect becomes especially pronounced at the high end of the temperature range, where prolonged exposure would encourage extensive phase transformations that undermine hardness gains.

Third, the discrepancy may stem from methodological differences in evaluating mechanical properties. Hardness alone does not capture the full spectrum of weld quality or the nuanced microstructural changes that occur at elevated temperatures. Past studies may have placed greater emphasis on other performance metrics—such as toughness, fatigue life, or ductility—leading them to recommend temperatures that did not align strictly with hardness optimization.

Lastly, differences in sample preparation and fixturing could have influenced the degree of stress relief and microstructural uniformity achieved. Even subtle variations in shielding configurations, sample geometry, or post-weld handling could produce outcomes that diverge from previously reported norms.

To resolve these issues, future investigations should employ more tightly controlled atmospheres, precisely timed heat cycles, and a broader suite of mechanical and microstructural characterization techniques. Doing so will clarify which variables most strongly influence hardness and help reconcile differences between this study's findings and established guidelines, thereby refining the criteria for selecting optimal post-weld heat treatment conditions for aerospace-grade titanium alloys.

6. Phase Transformations in Titanium Alloys

Hexagonal Close Packed HCP, α phase	Body Centred Cubic BCC, β phase	Phase transformations in titanium alloys are central to controlling their mechanical behavior and tailoring them to high-performance applications. The interplay between the hexagonal close-packed (HCP) alpha (α) phase and the body-centered cubic (BCC) beta (β) phase allows for a wide range of properties.
		

By manipulating these phases through alloying and thermal treatments, engineers can optimize strength, ductility, fatigue resistance, and corrosion resistance. Understanding these transformations is particularly critical in the context of welding and post-weld heat treatment (PWHT), where microstructural refinement can significantly enhance the final joint quality.

Pure titanium exhibits an allotropic transformation at approximately 882°C, where it changes from the α phase (HCP) stable at lower temperatures to the β phase (BCC) stable at higher temperatures. These two phases impart distinct property sets:

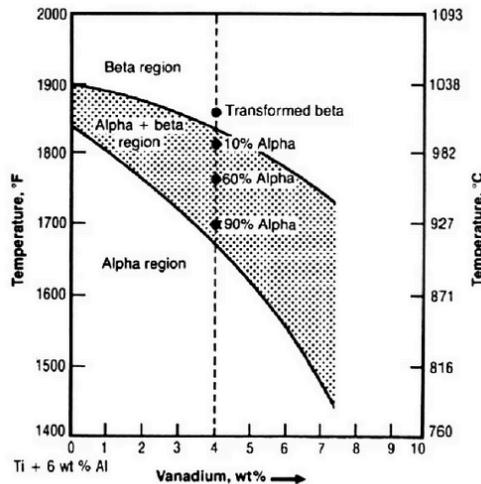
- **Alpha (α) Phase:** Higher strength, improved creep and corrosion resistance, generally lower ductility.
- **Beta (β) Phase:** More ductile, easier to form at elevated temperatures, but lower strength and creep resistance compared to α .

The temperature-dependent balance of these phases provides a flexible framework for microstructural engineering. By carefully selecting heat-treatment temperatures, it is possible to achieve a desired ratio of α to β phases, thus fine-tuning mechanical properties.

Alloying elements can stabilize or destabilize specific phases, effectively shifting the α/β transformation boundaries. For example:

- **Alpha Stabilizers (e.g., Al, Sn):** Raise the α -to- β transition temperature, expanding the temperature range in which α is stable.
- **Beta Stabilizers (e.g., V, Mo, Cr):** Lower the α -to- β transition temperature, enlarging the β field.
- **Neutral Elements (e.g., Zr):** Can strengthen both α and β phases.

- **Additions (e.g., Si):** Improve creep resistance or enhance other specialized properties.



Phase diagrams visually represent how phase fractions vary with temperature and

composition. They serve as a roadmap for understanding when and how titanium will transition between the α and β fields. For

Ti-6Al-4V, a widely used $\alpha+\beta$ alloy, phase diagrams show a two-phase region over a

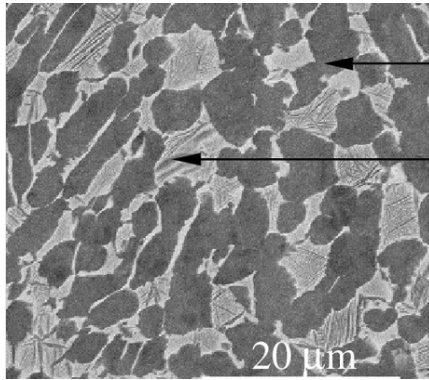
broad temperature range, allowing for targeted

thermal processing. Operating within the $\alpha+\beta$ region makes it possible to form refined microstructures that balance strength, ductility, and fatigue resistance.

Mechanical properties of Ti-6Al-4V are closely linked to the size, shape, and distribution of α and β grains. Fine, equiaxed α grains dispersed in a β matrix can yield a desirable combination of high strength, ductility, and corrosion resistance. On the other hand, coarse grains or a heavily skewed phase distribution may reduce fatigue life or toughness. By controlling the thermal cycle, including PWHT, it is possible to adjust the α -to- β ratio and the morphology of these phases, enhancing the weld's hardness and overall performance.

- **Fast Cooling (Quenching):** Can produce martensitic α' or fine α needles that increase strength but reduce ductility.
- **Moderate Cooling:** Maintains a balanced α/β mixture, often achieving a good compromise between strength and toughness.

- **Slow Cooling:** Encourages coarser α grains and retained β , improving ductility at the expense of some strength.



Aging and annealing steps further refine the microstructure after initial thermal treatments.

Aging encourages the precipitation of fine α particles within the β matrix, adjusting mechanical properties to meet specific performance goals.

Annealing can relieve residual stresses, enhance

stability, and improve fatigue life, making these processes critical tools for optimizing welded titanium components.

7. Conclusion

The results of this investigation highlight a notable deviation from established guidelines for post-weld heat treatment (PWHT) of Ti-6Al-4V. While prior recommendations suggested that optimal hardness would be achieved within a higher temperature range, our data showed that a significantly lower PWHT temperature of approximately 300°C yielded the greatest improvement in surface hardness. This unexpected outcome reflects the sensitivity of titanium's microstructure to subtle variations in environmental control, temperature holding times, and processing conditions.

Several factors likely contributed to these findings. First, the lack of an inert atmosphere during thermal treatment may have allowed oxygen ingress, promoting the formation of brittle surface layers at elevated temperatures and thus reducing measured hardness. Additionally, the unequal soak times associated with ramping up to higher

temperatures could have triggered extensive phase transformations that undermined strength and stiffness. These insights emphasize the importance of meticulous control over furnace atmosphere, heating rates, and hold times to achieve consistent and reproducible results.

Beyond the immediate mechanical properties, this work reinforces the complexity of titanium's $\alpha+\beta$ microstructure and the influential role of heat treatment in determining phase balance and morphology. Tuning the PWHT regime to achieve a more favorable distribution of phases can have profound implications for aerospace applications, where components must simultaneously meet stringent criteria for hardness, durability, and overall structural integrity.

Looking forward, future efforts would benefit from more controlled environments, enabling researchers to isolate the effects of temperature alone. Complementary techniques, such as microscopy or diffraction-based methods, could provide deeper insight into the microstructural changes driving the observed variations in hardness. Furthermore, expanding the range of mechanical tests beyond hardness measurements could offer a fuller picture of weld quality and component performance.

These findings highlight both the challenges and opportunities inherent in optimizing PWHT conditions for Ti-6Al-4V welds. Although our results deviate from conventional standards, they open the door to refining existing procedures and exploring welded titanium's mechanical properties in critical aerospace structures.

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